

Zachary J. Schroeder, PhD

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SUMMARY

Behavioral scientist with seven years of independently run research, I use language as data to study human behavior and perception (PhD, Psychology; specialization in educational data science). I specialize in integrating quantitative and qualitative research methods to turn messy data into clear insights. I have a passion for measurement (ensuring that the questions we ask get us the information we want) and data visualization for non-technical audiences.

EXPERIENCE

Research Scientist and Statistics Instructor (Graduate Research Fellow), University of Oregon, Eugene, OR Sep 2019 to Jun 2026

- Synthesized 20 years of video-recorded conversations into one analysis examining how empathy occurs in a conversation between two strangers. I used natural language processing (Python) and Bayesian multilevel Gaussian process models (R, Stan) to segment conversations into topics and sentences, code each sentence for its part of speech (question, suggestion, affirmation), and model similarity between what was said in the conversation and what people inferred their conversation partner was thinking. Contrary to previous research, I found that making suggestions was more reflective of someone's actual thoughts and feelings than asking questions.
- Independently built a Python and R text-analytics pipeline over 8M unstructured social-media posts (topic modeling, sentiment and affect scoring, engagement, network analysis). Knowing that social media algorithms tend to sort people into 'echo chambers,' I wanted to understand how members of different political groups discussed "cancel culture." I used semantic network analysis to identify partisan affiliation from each user's bio by modeling its similarity to other bios with explicitly partisan language. I then coded the moral and emotional content of each tweet using NLP classifiers and found that the two sides drew on different moral vocabularies (e.g., fairness, loyalty, degradation) to discuss the same cancellations.
- Designed and ran a three-study mixed-methods quasi-experimental program in Qualtrics (N = 1,125) on how gender and sense of belonging shape how people interpret ambiguous feedback: exploratory study, preregistered direct replication (power analysis, exclusion criteria, preregistered effect-size range for success), and preregistered extension; published in *Social Psychology of Education* (2025). Across the three studies, gender and sense of belonging predicted how positively students read the same ambiguous feedback, and more positive readings predicted choosing more challenging coursework.
- Modeled academic and disciplinary outcomes for 6,351 students across two national longitudinal datasets with preregistered multilevel logistic regressions and interaction contrasts. I found suspension disparities tracked adult judgment of student's personality more closely than students' actual (mis)behavior.
- Delivered 500+ hours of statistics instruction at the graduate and undergraduate level, including making interactive dashboards for students to explore statistical concepts (e.g., t-tests) and to develop familiarity with data science tools (e.g., SQL).

Executive Administrator, Medtronic, Mounds View, MN

May 2019 to Oct 2019

- Coordinated the merger of two departments (100+ employees), coordinated international travel for two senior VPs, and produced an all-staff meeting for 1,500 employees including content, filming, live stream, and materials.

SELECTED PORTFOLIO PROJECTS

Self-serve Shiny tools: [a step-by-step SQL tutorial on a live SQLite database](#), [an interactive t-test and ANOVA teaching app](#), and [a schedule builder for the LA28 Olympics](#) that uses budget, geographic location, and interest to create individualized schedules. More at zischroeder.com.

SKILLS

Experimentation and inference: A/B and factorial experiment design, power analysis, preregistration and replication, Bayesian multilevel and Gaussian process modeling, meta-analysis and pooled analysis across studies, regression with confounder adjustment and interaction contrasts, sensitivity analysis and cross-validation

Languages and tools: Python (pandas, scikit-learn, Hugging Face transformers, sentence-transformers), R (tidyverse, brms/Stan, lme4, ggplot2, Shiny, Quarto/R Markdown), data visualization (ggplot2, plotly, PowerBI), SQL (SQLite), Git/GitHub, SLURM/HPC, Qualtrics

Data and automation: pipelines over multi-million-row text corpora, dataset consolidation across sources, holdout validation of automated measures, self-serve Shiny dashboards and tools, reproducible automated reporting

Text and behavioral analytics: engagement modeling, sentence embeddings, dialogue-act and topic classification, sentiment and affect scoring (LIWC-22), qualitative coding at scale

EDUCATION AND PUBLICATIONS

PhD, Psychology, University of Oregon, April 2026. Specialization in Educational Data Science.

BA, Psychology (honors, cum laude) and Theatre, Northwestern University, June 2018.

Two peer-reviewed articles (*Social Psychology of Education*, 2025; *Frontiers in Psychology*, 2025); eight conference presentations.

[Google Scholar](#)